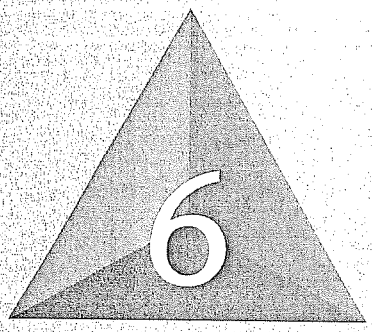
Experiment



Chemical Reactions of Copper and Percent Yield

OBJECTIVE

To gain some familiarity with basic laboratory procedures, some chemistry of a typical transition element, and the concept of percent yield.

APPARATUS AND CHEMICALS

Apparatus

balance
250-mL beakers (2)
evaporating dish
stirring rod
towel
wire gauze
Bunsen burner and hose
100-mL graduated cylinder
weighing paper
boiling chips
ring stand and iron ring

Chemicals

0.5-g piece of copper wire (16 or 18 gauge)
6 *M* H_2SO_4
~~methanol~~ ethanol
aluminum foil cut in 1-in. squares
conc. HNO_3
3.0 *M* NaOH
granular zinc
acetone
conc. HCl

DISCUSSION

Most chemical syntheses involve the separation and purification of the desired product from unwanted contaminants. Common methods of separation are filtration, sedimentation, decantation, extraction, and sublimation. This experiment is designed to be a quantitative evaluation of your individual laboratory skills in carrying out some of these operations. At the same time, you will become acquainted with two fundamental types of chemical reactions, called redox reactions and metathesis reactions. By means of these reactions, you will carry out several chemical transformations involving copper and its compounds, and finally you will recover the copper sample with maximum efficiency. The chemical reactions involved are the following:

$$Cu(s) + 4HNO_3(aq) \longrightarrow Cu(NO_3)_2(aq) + 2NO_2(g) + 2H_2O(l)$$ Redox [1]

$$Cu(NO_3)_2(aq) + 2NaOH(aq) \longrightarrow Cu(OH)_2(s) + 2NaNO_3(aq)$$ Metathesis [2]

$$Cu(OH)_2(s) \xrightarrow{\Delta} CuO(s) + H_2O(g)$$ Dehydration [3]

$$CuO(s) + H_2SO_4(aq) \longrightarrow CuSO_4(aq) + H_2O(l)$$ Metathesis [4]

$$CuSO_4(aq) + Zn(s) \longrightarrow ZnSO_4(aq) + Cu(s)$$ Redox [5]

$$3CuSO_4(aq) + 2Al(s) \longrightarrow Al_2(SO_4)_3(aq) + 3Cu(s)$$ Redox [6]

Each of these reactions proceeds to completion. Metathesis reactions proceed to completion whenever one of the products is removed from the solution, such as in the formation of a gas or an insoluble substance.

This is the case for reactions [1], [2], and [3], where in reactions [1] and [3] a gas and in reaction [2] an insoluble precipitate are formed. (Reactions [5] and [6] proceed to completion because copper is more difficult to oxidize than either zinc or aluminum.)

The object of this experiment is to recover all of the copper with which you began. This is the test of your laboratory skills.

The percent yield of the copper can be expressed as the ratio of the recovered mass to initial mass, multipled by 100:

$$\% \text{ yield} = \frac{\text{recovered mass of Cu}}{\text{initial mass of Cu}} \times 100$$

PROCEDURE

Weigh approximately 0.500 g of no. 16 or no. 18 copper wire to the nearest 0.0001 g and record its mass (1). Place it in a 250-mL beaker. IN THE HOOD add 4 or 5 mL of concentrated HNO_3 to the beaker. (**CAUTION:** ***Be careful not to get any of the nitric acid on yourself. If you do, wash it off immediately with copious amounts of water. The gas produced in this reaction is toxic, and the reaction must be performed in the hood.***) After the reaction is complete, add 100 mL of distilled H_2O. Describe the reaction as to color change, evolution of a gas, and change in temperature (exothermic or endothermic) on the report sheet (6).

Add 30 mL of 3.0 *M* NaOH to the solution in your beaker and describe the reaction on the report sheet (7). ~~Add two or three boiling chips and~~ carefully heat the solution—while stirring with a stirring rod—just to the boiling point. ~~Describe the reaction on your report sheet (8).~~

Allow the black CuO to settle, then decant the supernatant liquid. Add about 200 mL of very hot distilled water, stir, and then allow the CuO to settle. Decant once more. What are you removing by the washing and decantation (9)?

Add 15 mL of 6.0 *M* H_2SO_4. What copper compound is present in the beaker now (10)?

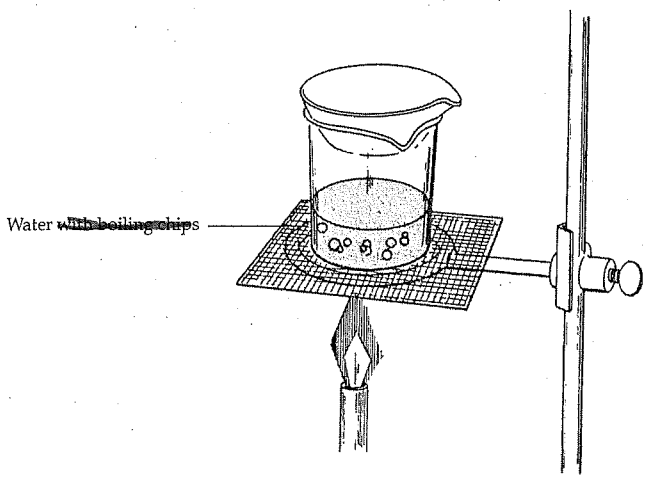
Your instructor will tell you whether you should use zinc or aluminum for the reduction of Cu(II) in the following step.

~~A.~~ Reduction with Zinc

In the hood, add ~~2.0 g of~~ 30-mesh zinc metal all at once and stir ~~until the~~ supernatant liquid is colorless. ~~Describe the reaction on your~~ report sheet (11). What is present in solution (12)? ~~When gas evolution has~~ become *very* slow, heat the solution ~~gently (but do not boil) and allow it to cool. What gas~~ is ~~formed in this~~ reaction (13)? How do you know (14)?

B. Reduction with Aluminum

In the hood, add several 1-in. squares of aluminum foil and a few drops of concentrated HCl. Continue to add pieces of aluminum until the supernatant liquid is colorless. Describe the reaction on your report sheet (11). What is present in solution (12)? What gas is formed in this reaction (13)? How do you know (14)?



Do not use bunsen burner, use hot plate

▲ **FIGURE 6.1** Steam bath.

For Either Reduction Method When gas evolution has ceased, decant the solution and transfer the precipitate to a preweighed porcelain evaporating dish and record its mass on the report sheet (3). Wash the precipitated copper with about 5 mL of distilled water; allow it to settle, decant the solution, and repeat the process. What are you removing by washing (15)? Wash the precipitate with about 5 mL of ~~methanol~~. **(CAUTION:** ***Keep the ~~methanol~~ away from flames—it is flammable!! Methanol is also extremely toxic. Avoid breathing the vapors as much as possible.)*** Allow the precipitate to settle, and decant the ~~methanol~~. Finally, wash the precipitate with about 5 mL of acetone. **(CAUTION:** ***Keep the acetone away from flames—it is extremely flammable!!)*** Allow the precipitate to settle, and decant the acetone from the precipitate. Prepare a steam bath as illustrated in Figure 6.1 and dry the product on your steam bath for at least 5 min. Wipe the bottom of the evaporating dish with a towel, remove the boiling chips, weigh the evaporating dish plus copper, and record its mass (2). Calculate the final mass of copper (4). Compare the mass with your initial mass and calculate the percent yield (5). What color is your copper sample (16)? Is it uniform in appearance (17)? Suggest possible sources of error in this experiment (18).

use ethanol, not Methanol

use hot plate

Dispose of the chemicals in the designated receptacles.

PRE LAB QUESTIONS

Before beginning this experiment in the laboratory, you should be able to answer the following questions:

1. Give an example, other than the ones listed in this experiment, of redox and methathesis reactions.
2. When will reactions proceed to completion?
3. Define *percent yield* in general terms.
4. Name six methods of separating materials.
5. Give criteria in terms of temperature changes for exothermic and endothermic reactions.
6. If 3.35 g of $Cu(NO_3)_2$ are obtained from allowing 2.25 g of Cu to react with excess HNO_3, what is the percent yield of the reaction?

7. What is the maximum percent yield in any reaction?
8. What is meant by the terms *decantation* and *filtration*?
9. When $Cu(OH)_2(s)$ is heated, copper(II) oxide and water are formed. Write a balanced equation for the reaction.
10. When sulfuric acid and copper(II) oxide are allowed to react, copper(II) sulfate and water are formed. Write a balanced equation for this reaction.